

CNN (Convolutional Neural Network).

07/08/20

* These are differentiated from other neural networks by their superior performance with image, speech or audio signal inputs.

→ Three main layers

* Convolutional Layer

* Pooling Layer

* fully connected layer

Convolutional Layer

* This is the first layer that is used to extract the data.

* kernels perform element-wise multiplication & results the summation of it, which is called as feature map.

* kernels are small.

* common dimensions include 3×3 , 5×5 , 7×7 .

$$a_1 \ a_2 \ a_3 \ a_4 \ a_5$$

b_1	1	1	1	0	0
b_2	0	1	1	1	0
b_3	0	0	1	1	1
b_4	0	0	1	1	0
b_5	0	1	1	0	0

Input

1	0	1
0	1	0
1	0	1

Filter

* multiply all the ~~main~~ elements with the given filter.

→ First $(a_1, a_3) \& (b_1, b_3) \times \text{Filter}$

→ $(a_2, a_4) \& (b_1, b_3) \times \text{Filter}$

→ $(a_3, a_5) \& (b_1, b_3) \times \text{Filter}$

(a_1, a_3)	(a_2, a_4)	(a_3, a_5)
(b_1, b_3)	(b_1, b_3)	(b_1, b_3)
(a_1, a_3)	(a_2, a_4)	(a_3, a_5)
(b_1, b_3)	(b_1, b_3)	(b_1, b_3)
(a_1, a_3)	(a_2, a_4)	(a_3, a_5)
(b_1, b_3)	(b_1, b_3)	(b_1, b_3)

Feature Map

$$W_{out} = \frac{W - F + 2P}{S} + 1$$

Stride & Padding

W = Input size.

F = Filter size

P = Padding

S = Stride

Ex:- $W \times H \times D$
 $28 \times 28 \times 3$

$$K = 16$$

$$F = 5$$

$$S = 1$$

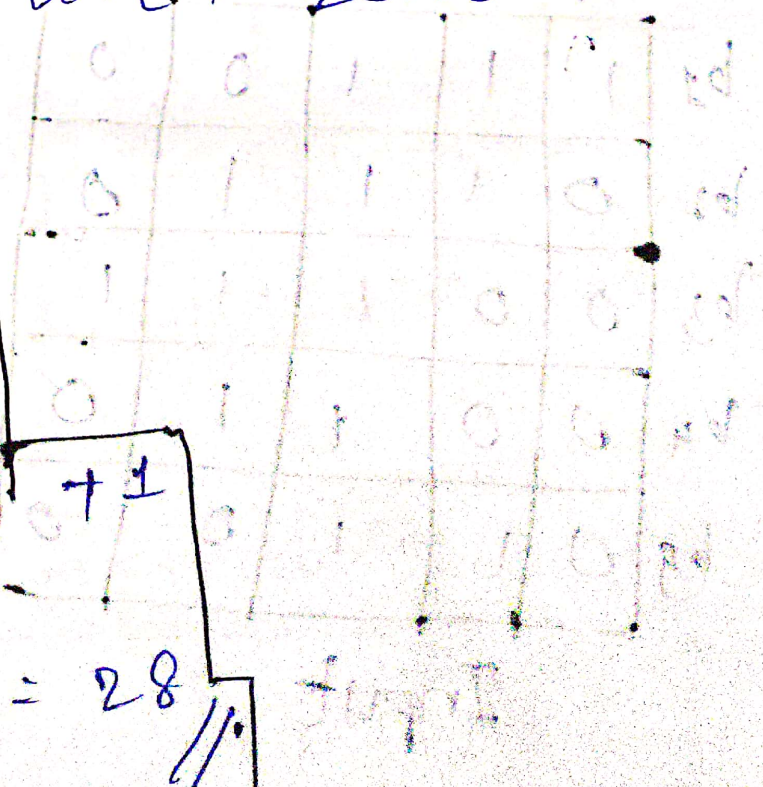
$$P = 2$$

$$W = \frac{(28 - 5 + 4)}{1} + 1$$

$$= 27 + 1 = 28 //$$

* Stride specifies how much we move the convolution filter at each step

* To maintain the same dimensionality we can use padding to surround the input with zeros.



random sample.

$$\theta = \theta - \eta \nabla L_i(\theta)$$

Softmax

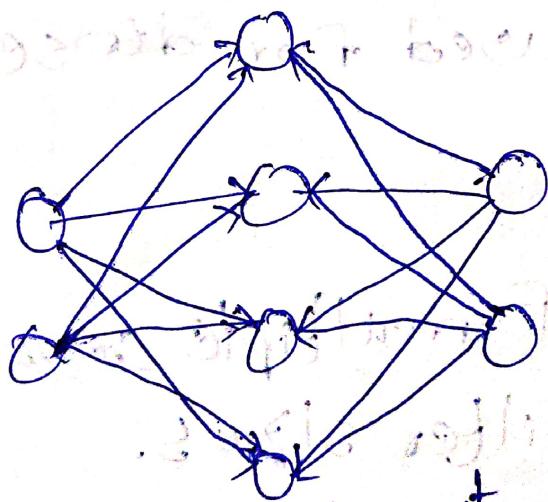
* It converts the output of the last layer in neural network into a probability distribution.

* Mainly used to normalize neural networks output to fit b/w zero & one.

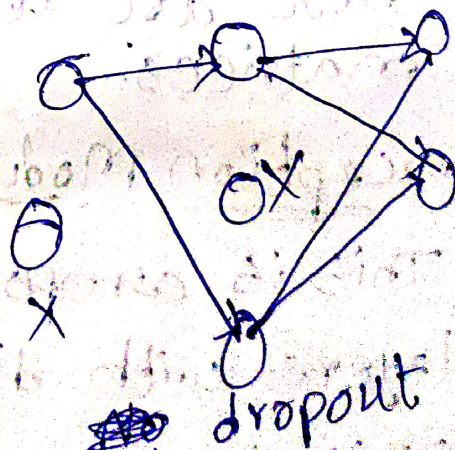
Dropout

* This is used to avoid overfitting.

* By using dropout we can increase the accuracy.



without dropout



with

dropout

Typical dropout rate = 0.2 to 0.5

Difference between AlexNet & ZFnet 3/9/2018

AlexNet

- * 8 layer (5 convolutional layer, 3 fully connected layers).
- * 11×11 , 5×5 , 3×3 filter sizes.
- * Stride of 4 in the first layer
- * Local Response Normalization

ZFnet

- * This had some architectural adjustments compared to ZFnet.
- * 7×7 (Filter size in first layer) & 3×3 in the latter layer
- * Stride of 2 is used in the first layer.
- * Local Contrast Normalization

GoogleNet (Inception-v1)

- * In this multiple convolutional layers are combined.
- * These are better used for dense matrices.

Inception module

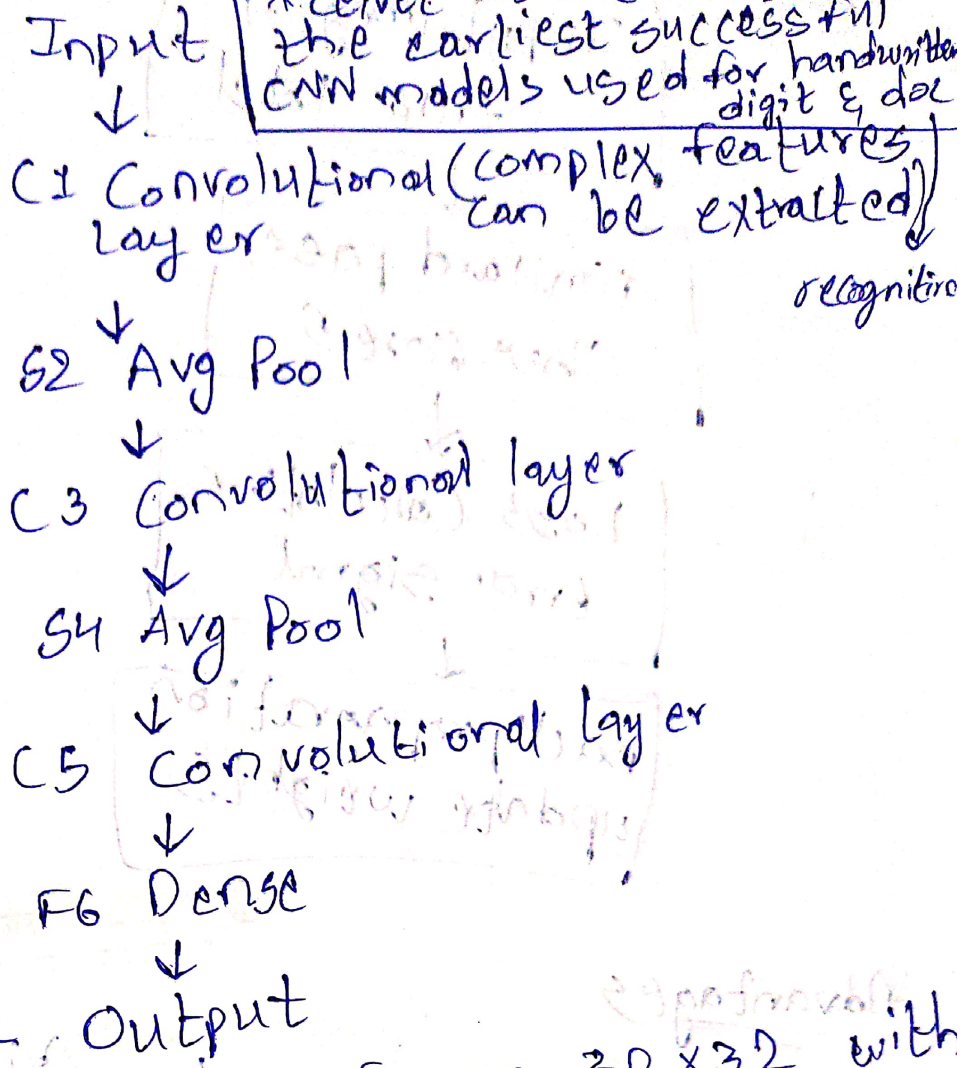
- * This is composed of multiple convolutional layers with diff filter sizes.
- * This is computationally expensive & this can be avoided by adding the 1×1 convolutional layer before the convolutional layers.

LENET-5

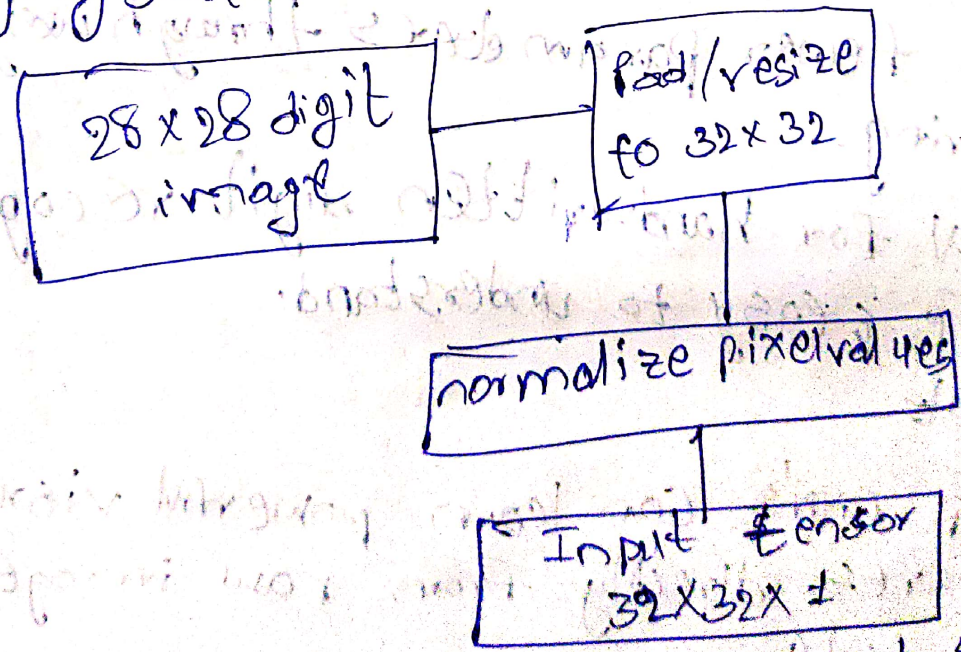
19/08/25

* LeNet is one of the earliest successful CNN models used for handwritten digit & doc recognition

Architecture of LENET-5

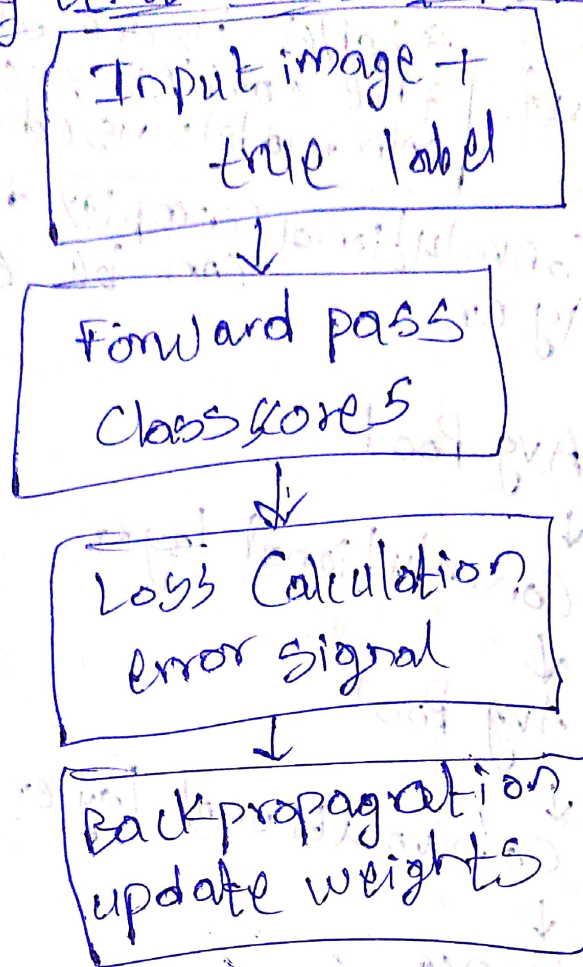


* Input image is of size 32×32 with gray scale.



* Pixel values are normalized so that training is stable.

Training LeNet with Backpropagation



Advantages

- * Learns features automatically for pixels.
- * Uses fewer parameters through weight sharing.
- * Useful for handwritten digit recognition.
- * Simple & easy to understand.

AlexNet

- * Deep CNN's can learn powerful visual features directly from raw images at large scale.
- * This is the bridge b/w basic CNN theory & modern deep visual recognition systems.

* Handcrafted features

Architecture

227x227x3
RGB

* ReLU Activation
helped AlexNet
train deeper
models faster
than saturating
activations.

Input
↓
Conv1
↓
Pool
↓
Conv2
↓
Conv3
↓
Conv5
↓
FC6-FC7
↓
Softmax

LeNet
(small CNN)

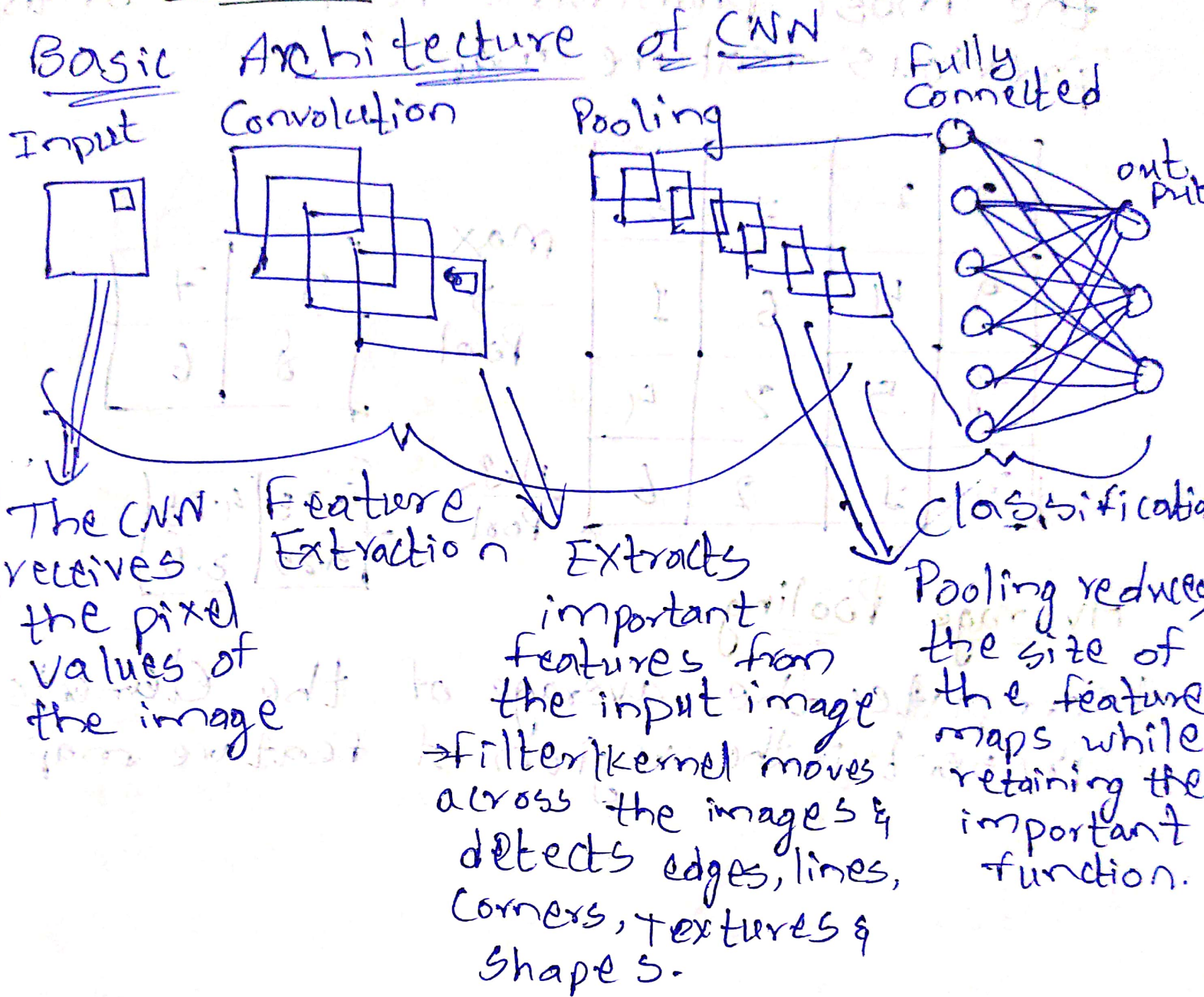
AlexNet
(deep CNN)

Modern CNNs
(VGG)

~~Architecture~~

* Designed for self

Convolutional Layer



Pooling Layer

* These are used to reduce the dimensions of the feature maps.

* Reduces the number of parameters to learn & the amount of computation performed in the network.

$$W_{out} = \frac{W - F}{S} + 1$$

Max Pooling

* This is a pooling operation that selects the max element from the region of the feature map covered by the filter.

* This gives the feature map containing the most prominent features of the previous feature map.

2	2	7	3
9	4	6	1
8	5	2	4
3	1	2	6

max
→
Pool

9	7
8	6

Avg
→
Pool

4.25	4.25
4.25	3.5

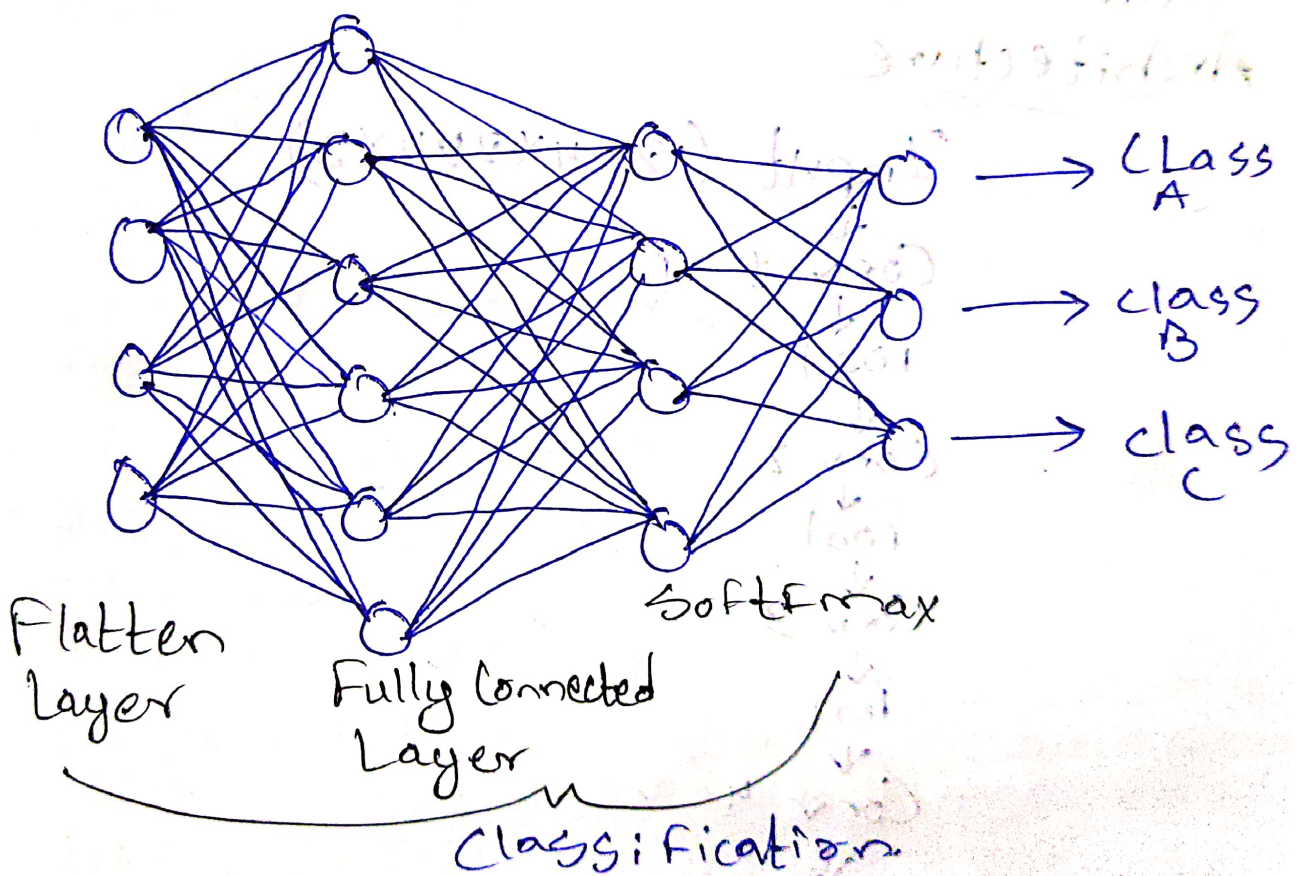
Average Pooling

* Computes the average of the elements present in the region of feature map.

Fully Connected Layer

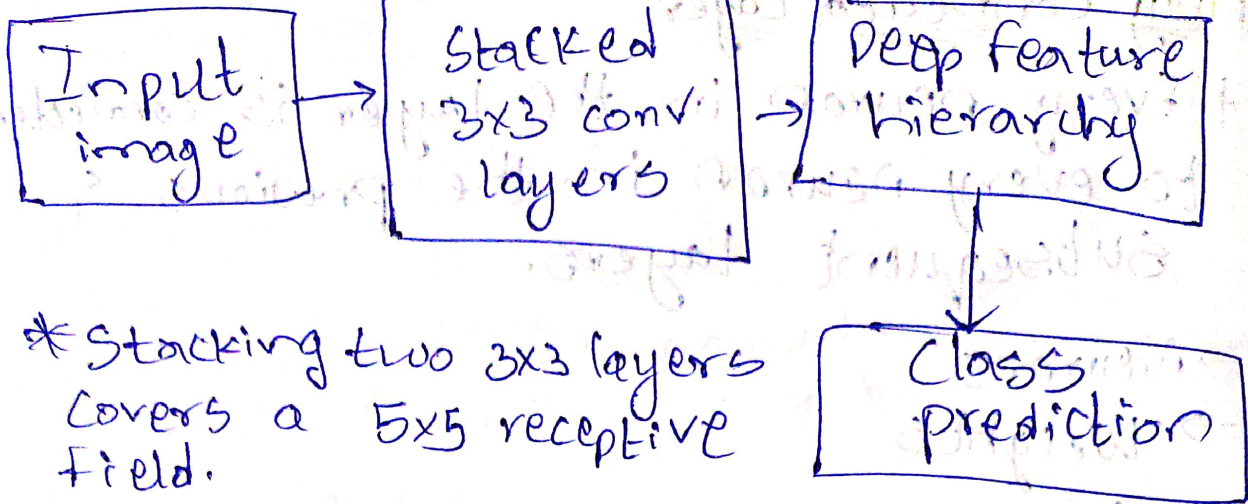
* Every neuron in the layer is connected to every neuron in the previous & subsequent layers.

- Neurons
- Weights
- Biases
- Activation Function



VGG Net

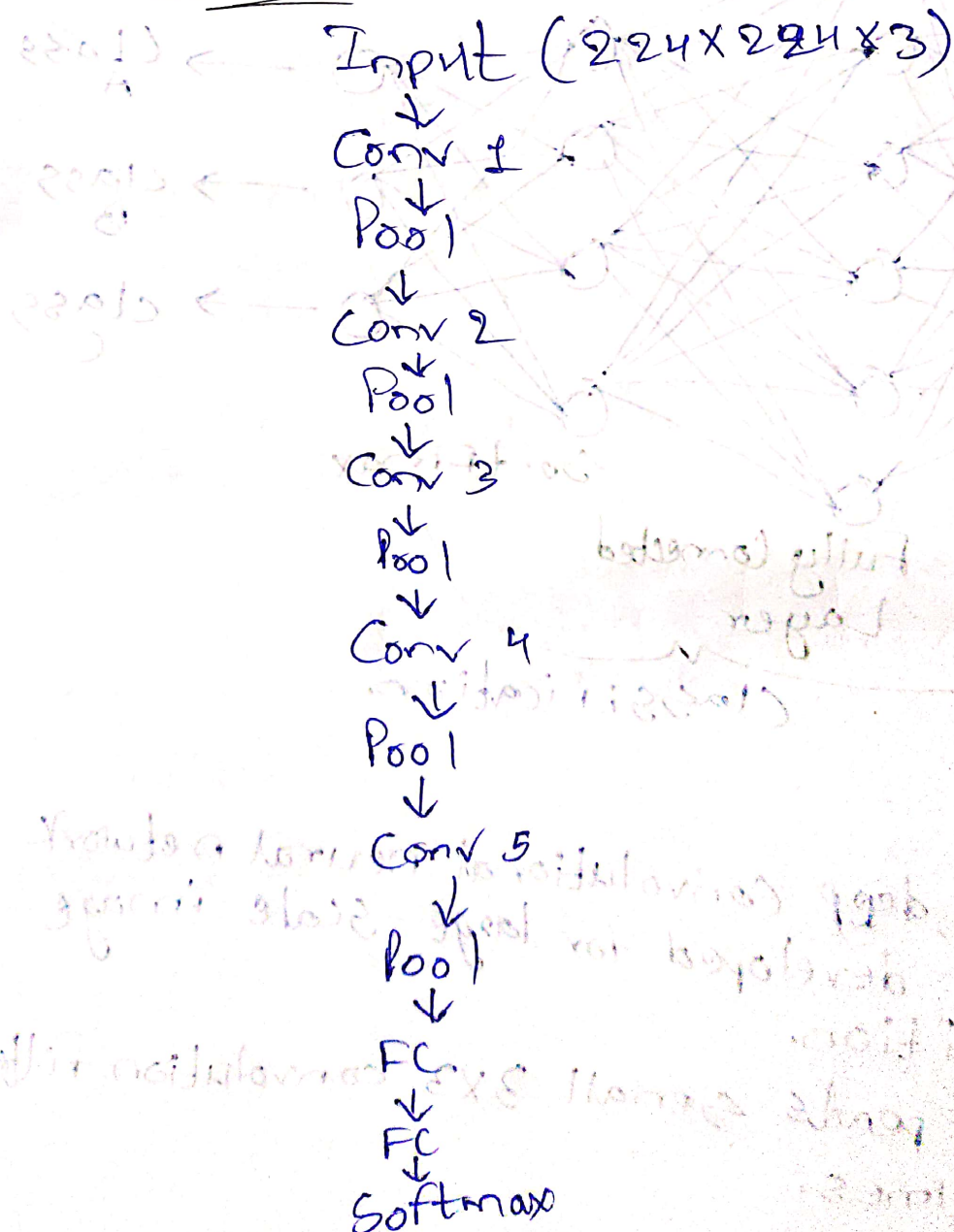
- * A very deep convolutional neural network family developed for large-scale image recognition.
- * This repeats small 3×3 convolution filter many times.



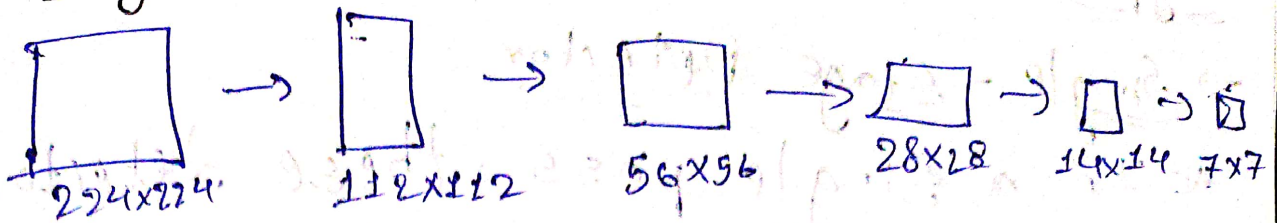
* Stacking two 3x3 layers covers a 5x5 receptive field.

* Stacking three 3x3 layers covers a 7x7 receptive field.

Architecture



pooling



* Pooling reduces computation, by shrinking spatial dimensions.

Advantages

* Strong feature extractor for transfer learning.

* Shows clear benefit of increasing CNN depth.

→ VGGNet is the best bridge bwn early CNN's & modern deep architectures.

Applications

→ Image Classification

→ Feature extraction.

→ Transfer Learning.

→ Medical image analysis.

→ Content & Style features.

Object Detection

* Object detection is a computer vision task that identifies & locates objects within an image.

Types

→ Single - Stage Detector

- * In a single pass, these detectors accurately forecast the bounding boxes and class probabilities for every area of picture.

Ex:- SSD, YOLO.

→ Two - Stage Detector

- * First a candidate region is proposed.
- * Then the region is classified into categories.

Ex:- R-CNN, Fast R-CNN, Faster R-CNN.

RCNN

- * Region - Based CNN.

* Uses selective search to generate around 2000 region proposals, resizes each region, extract features with a pre-trained CNN.

- * Maintains good accuracy.

Fast R-CNN

- * Faster ~~er~~ R-CNN.

↓
than.

- * Uses selective search.

Faster R-CNN

* Even faster than R-CNN.

* The major problem in faster R-CNN is it still uses selective search.